

Sample University Syllabus

Program Information

Program: B.Tech Artificial Intelligence and Data Science

Duration: 4 Years / 8 Semesters

Academic Year: 2026–2027

Semester 5 – Core Subjects

- Machine Learning – supervised learning, unsupervised learning, model evaluation and feature engineering.
- Deep Learning – neural networks, CNNs, RNNs, optimization and model training.
- Big Data Analytics – Hadoop, HDFS, MapReduce and distributed data processing.
- Database Management Systems – SQL, normalization, transactions and indexing.
- Data Visualization – Matplotlib, dashboards, statistical plots and visual analytics.

Practical Components

- Machine Learning Laboratory
- Database Laboratory
- Mini Project
- Technical Seminar

Assessment Pattern

Internal assessment: 40%. End-semester examination: 60%. Laboratory courses are evaluated through continuous practical work and a final practical examination.

Learning Outcomes

- Apply machine learning algorithms to real-world datasets.
- Develop and evaluate data-driven applications.
- Understand distributed data processing concepts.
- Communicate technical results using appropriate visualizations.